The Signaling Gap

State-Owned Media and the Strategic Management of Sentiment during the Russian Invasion of Ukraine

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Abstract

Alliance commitments enable cooperation but can foreclose dissent: a state bound to a patron cannot openly break with it without risking its security guarantee. We argue state-owned media is an institutionally mediated channel — public enough for the foreign governments that monitor it to read, but not a formal act of state.

Across 174,094 articles from 11 countries around the 2022 invasion of Ukraine, states with Russian security agreements shifted **more negative** toward Russia than neutral states — even while refusing to condemn it through formal channels. We measure sentiment with a prompted large language model and identify the shift with a regression discontinuity.

The Argument

The bind. A state dependent on a patron cannot openly dissent without risking the guarantee it relies on — yet silence risks being lumped in with the patron.

Three channels, one gap.

- Formal channels (UN votes) are credible but costly — unusable under a patron's threat.
- Private diplomacy is safe but unverifiable — cheap talk.
- State media sits between: observable, but not a formal act of state.

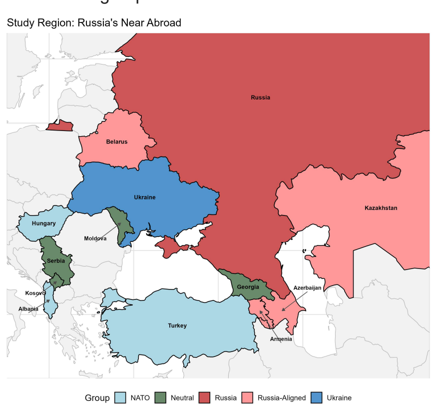
Why it's credible. Readable (long read as regime preference), costly (negative coverage risks provoking the patron), and buffered (editorial separation keeps it below a formal break).

The calculus. Signaling risks retaliation\; silence risks being categorized with the aggressor. However, the higher costs associated with diplomatic isolation makes sending the signal rational, and the logic should travel to any patron–client hierarchy.

The Case: Why the 2022 Invasion

A clean test needs three things, and February 24, 2022 supplies all three:

- **A discrete shock** — a sharp rupture in the information environment, not a slow slide.
- **A constraint structure** — states bound to Russia (CSTO + Azerbaijan) vs. neutral states with open channels.
- **Competing predictions** — domestic-management and signaling disagree on which group shifts more.



Hypotheses

H1 · Baseline shock — the invasion drove a region-wide negative shift toward Russia.
H2 · Readability — state media manages coverage more carefully than independent media.
H3 · The signal — Russia-aligned state media shifts more negative than neutral state media.

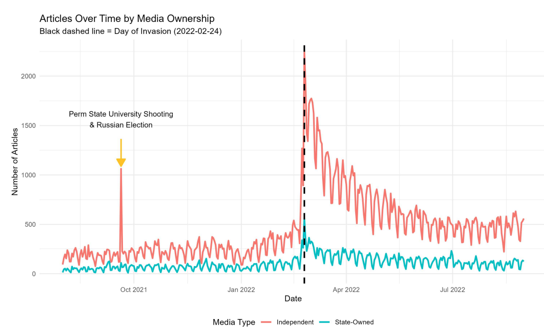
H3 pits the two accounts head-to-head:

Domestic management	Signaling (our account)
Suppress coverage → **predicts less negative**	Signal via media → **predicts more negative**

Data: shifts **more negative** ✓ **signaling, not suppression**

Data & Measurement

- **174,094** Russia-related articles, Aug 2021–Aug 2022; 11 countries, 61 sources (17 state-owned, 44 independent).
- **GPT-4o** codes sentiment toward Russia (−1 / 0 / +1) from each title and first two sentences. F1 = 0.92 vs. a 100-article human standard — with no −1↔+1 cross-codings, a conservative measure.



Daily Russia-related articles by ownership; dashed line marks the Feb 24 invasion.

Research Design

H1 (within group). An RD at Feb 24 identifies the causal effect of the invasion on sentiment (rdrobust, MSE-optimal bandwidth).
H2 / H3 (across groups). A difference-in-discontinuities (RDID) compares how much more one group jumped at the cutoff:

$$Y_p^z = \beta_0 + \beta_1 D_p + \beta_2 T_p + \beta_3 D_p T_p + \beta_4 G_p + \beta_5 T_p G_p + \beta_6 D_p G_p + \beta_7 D_p T_p G_p + \gamma_{\text{topic}} + \varepsilon_p$$

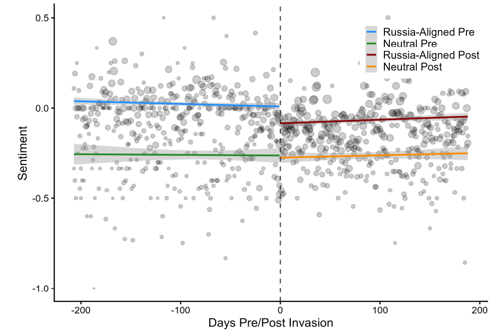
β5 (T × G) is the differential discontinuity — causal within each group, but *conditional on group membership* across groups (stress-tested, not a causal effect of alignment).
Identification. Source-level Z-Score adjustments close country/ownership backdoors; topic FE absorb composition; WCR bootstrap handles the few clusters.

Results

H1 — A real shock. Sentiment toward Russia fell 0.22 SD at the cutoff ($p < 0.001$); 0.18 SD without military coverage.

H2 — State media holds the line. Independent media fell 0.21 SD; state-owned only 0.11 SD. RDID interaction 0.22 ($p < 0.05$) among non-NATO outlets.

H3 — Constrained states signaled loudest. Russia-aligned state media fell −0.22 SD ($p < 0.05$); neutral barely moved (−0.06, n.s.). Matched estimate −0.38 ($p<0.05$) — the opposite of domestic management prediction



Russia-aligned drops sharply at the cutoff; neutral stays flat.

In the outlet's own words. Armenia's state broadcaster, two months apart — January: Russia as an economic partner; March: 'Russia walks out of the Council of Europe.' Same outlet, opposite orientation — while Armenia withheld formal condemnation.

Robustness

Sensitivity (Cinelli–Hazlett). A confounder would need to be 1.5× as strongly associated with sentiment as the invasion itself to erase the H3 differential.
Placebo + ethnic partition. No discontinuity at the Dec 7 announcement; splitting by ethnic-Russian share yields no differential.

Conclusion

- Connects patron–client theory and IR signaling — constrained clients use managed media to signal divergence from a patron.
- Controlled-media sentiment is a deliberate signal of alignment toward a security partner — not a byproduct of domestic opinion.
- Validates a long-standing intelligence practice — reading state media for regime preferences — long left unexamined.

Key References

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